

Full Stack Development Summary

File Handling

File handling in java allows programs to create , read , write , append, and delete files stored on disk. Java provides classes for handling files mainly inside:

Java.io.package

- ✚File
- ✚FileReader, BufferedReader
- ✚FileWriter, BufferWriter
- ✚FileInputStream, FileOutputStream

Why use BufferedWriter

BufferedWriter writes text to a file faster by collecting data in a buffer before writing it all at once instead of writing character by character.

Why use BufferedWriter?

Without BufferedWriter

Writes directly to file each time

Slower for large content

More disk operations

With BufferedWriter

Stores data in memory first

Faster and more efficient

Fewer disk operations

Key Features

- ✚ Speeds up writing large text file
- ✚ Support `newLine()` method to easily add new lines.
- ✚ Work together with `FileWriter, FileOutputStream,`

Remember

- ✚ `FileWriter` alone can write text
- ✚ `BufferedWrite` improves performance by buffering the data

Sample Program

```

package practice_java;

import java.io.FileNotFoundException;
import java.io.FileReader;

public class Mainclass {

    public static void main(String[] args) {

        try {
            FileReader fr = new FileReader("output.txt");
            int c = fr.read();
            System.out.println((char)c);
            c = fr.read();
            System.out.println((char)c);
            fr.close();
        }
        catch (Exception e) {

        }

    }
}

```

Buffer Reader in Java

BufferedReader is a class in Java used to read text from a file (or input source) efficiently by using a buffer (temporary memory area).

Why use BufferedReader?

Without BufferedReader

Reads character-by-character

Slower for large files

More disk operations

With BufferedReader

Reads big chunks of text at once

Faster and more efficient

Fewer disk operations

Sample program

```
import java.io.FileNotFoundException;
import java.io.FileReader;
import java.io.BufferedReader;

public class Mainclass {

    public static void main(String[] args) {

        try {
            FileReader fr = new FileReader("output.txt");
            BufferedReader br = new BufferedReader(fr);

            String line = br.readLine();

            while (line != null) {
                System.out.println(line);
                line = br.readLine();
            }

            br.close();
        }
        catch (Exception e) {
        }
    }
}
```